Homework 3

Due: *Monday, June 7th at 11:59pm EDT*

All homeworks are due the Monday after their release at 11:59pm EDT on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit. We recommend checking out the sample proofs on the course website to see the level of rigor for which we're looking.

Problem 1

For this problem, let

$$\begin{aligned} A &= \{1, 2, 3, 4, 5\} \\ B &= \mathcal{P}(\{1, 4, 9\}) \\ C &= \{x \mid \exists\, y \in \mathbb{Z} \text{ such that } y^2 = x,\ x < 10\} \end{aligned}$$

a. Find the following sets:

   i. $A \cup C$
   ii. $A \cap B$
   iii. $A - C$
   iv. $C \times C$
   v. $\{x \mid x \in C \text{ AND } \exists\, y, z \in A \text{ such that } y + z = x\}$

b. Find the cardinalities of the following sets:

   i. $C \times C$
   ii. $\mathcal{P}(C \times C)$
   iii. $\mathcal{P}(\mathcal{P}(A) \cup B)$

c. Assume that the sets A, B, C are all subsets of the same universal set U. **Using set algebra**, show that the following equivalences hold.

   i. $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
   ii. $A \cup (B \cap (C - A)) = A \cup (B \cap C)$
   iii. $\overline{(A \cup (B \cap C))} = (\overline{C} \cup \overline{B}) \cap \overline{A}$

Problem 2

Consider a finite set S and its power set, $\mathcal{P}(S)$. For parts a through d, find and prove the size of the relevant set. If it is non-empty, state which elements of $\mathcal{P}(S)$ are in it and explain why.

a. $K_1 = \{x \in \mathcal{P}(S) \mid \forall y \in \mathcal{P}(S),\ x \cap y = \emptyset\}$.

b. $K_2 = \{x \in \mathcal{P}(S) \mid \forall y \in \mathcal{P}(S),\ x \cap y = x\}$.

c. $K_3 = \{x \in \mathcal{P}(S) \mid \forall y \in \mathcal{P}(S),\ x \cap y = y\}$.

d. $K_4 = \{x \in \mathcal{P}(S) \mid \forall y \in \mathcal{P}(S),\ x \cup y = x\}$.

e. What is the intersection of K_1, K_2, K_3, and K_4?

f. What is the union of K_1, K_2, K_3, and K_4?

Problem 3

A binary operation, $\star$, is an operation on a set that takes in two elements from the set and returns a third. For example, the addition operation $+$ is a binary operation on the integers.

We say that a binary operation $\star$ on a set S is *commutative* if

$$x \star y = y \star x \text{ for all } x, y \in S.$$

Let X be a finite set. For each of the following operations, prove whether or not the operation is commutative over $\mathcal{P}(X)$.

i. Set union

ii. Set intersection

iii. Set difference

iv. Symmetric difference

Mind Bender (Extra Credit)

Mad about the chaos that ensued from the game of asteroid gravitational deflection, Venus is looking to instead host the parties for all future syzygies. It looks to invite distinct groups of celestial bodies to each one. To do so, it begins writing out the powerset of the set of celestial bodies in the solar system.

Bored of how surprisingly long it's taking, Venus starts to think about different properties of the powerset. It considers the powerset of the powerset of a set A. If its first element is the set that consists of the set A, the second element is the powerset of A, the third element is A's only element, and the fourth and last element is A's only element's only element, then what is A? Help Venus answer this question to get back to planning.

Note: Make sure to prove your answer is correct *and* unique.